# Daily Problem: 09–Feb–2026

## Problem Statement

Construct a weighted automaton over the real numbers with a single-symbol alphabet $\{a\}$ that assigns weight

$$(1-\alpha)\alpha^k$$

to the string $a^k$, for a parameter $\alpha$.

Now consider the generalisation $D(\alpha, m, c)$: it assigns weight $(1-\alpha)\alpha^k$ to strings of the form $a^{mk+c}$, and weight 0 to all other strings (i.e. strings whose length is not congruent to $c \bmod m$). Show that $D(\alpha, m, c)$ can also be represented by a weighted automaton.

## Weighted Automata Model

A (real) weighted automaton is a tuple

$$A = (Q, \Sigma, \mu, \lambda, \rho)$$

where $Q$ is a finite state set, $\Sigma$ is the alphabet, $\lambda \in \mathbb{R}^{1\times|Q|}$ is the initial weight row vector, $\rho \in \mathbb{R}^{|Q|\times 1}$ is the final weight column vector, and $\mu : \Sigma \to \mathbb{R}^{|Q|\times|Q|}$ maps each symbol to a transition-weight matrix.

For a word $w = x_1 x_2 \cdots x_n$, the weight assigned is

$$\mathrm{wt}(w) = \lambda\, \mu(x_1)\mu(x_2)\cdots\mu(x_n)\, \rho.$$

For the empty word $\varepsilon$, $\mathrm{wt}(\varepsilon) = \lambda\rho$.

## Part 1: Geometric Distribution on $\{a\}^*$

We want $\mathrm{wt}(a^k) = (1-\alpha)\alpha^k$ for all $k \geq 0$.

### Construction

Take a one-state weighted automaton $A_\alpha$ with $Q = \{q\}$ (so all matrices are $1 \times 1$). Define:

$$\lambda = \begin{bmatrix} 1 \end{bmatrix}, \qquad \mu(a) = \begin{bmatrix} \alpha \end{bmatrix}, \qquad \rho = \begin{bmatrix} 1-\alpha \end{bmatrix}.$$

### Verification

For $k \geq 0$,

$$\mathrm{wt}(a^k) = \lambda\, \mu(a)^k\, \rho = \begin{bmatrix} 1 \end{bmatrix} \begin{bmatrix} \alpha \end{bmatrix}^k \begin{bmatrix} 1-\alpha \end{bmatrix} = (1-\alpha)\alpha^k.$$

So the automaton generates exactly the geometric distribution on lengths.

(If we want it to be a probability distribution, we assume $0 \leq \alpha < 1$, in which case $\sum_{k \geq 0} (1-\alpha)\alpha^k = 1$.)

## Part 2: The Generalised Distribution $D(\alpha, m, c)$

Fix integers $m \geq 1$ and $c \in \{0, 1, \ldots, m-1\}$. The distribution $D(\alpha, m, c)$ assigns:

$$\mathrm{wt}(a^n) = \begin{cases} (1-\alpha)\alpha^k & \text{if } n = mk + c \text{ for some } k \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Equivalently, the weight is nonzero exactly when $n \equiv c \pmod{m}$, and then depends only on $k = (n-c)/m$.

### Idea

We need:

- a "mod-$m$ counter" to ensure the length is congruent to $c$,
- a factor of $\alpha$ for each full block of $m$ symbols after the first $c$ symbols,
- and a final factor $(1-\alpha)$ exactly at acceptance.

## Construction

Let the state set be

$$Q = \{0, 1, \ldots, m-1\},$$

where state $i$ represents the current length modulo $m$.

Define the transition matrix $\mu(a) \in \mathbb{R}^{m\times m}$ by:

$$\mu(a)_{i,j} = \begin{cases} 1 & \text{if } j \equiv i+1 \pmod{m} \text{ and } i \neq m-1, \\ \alpha & \text{if } i = m-1 \text{ and } j = 0, \\ 0 & \text{otherwise.} \end{cases}$$

In words: reading $a$ advances the residue by 1; when we wrap from $m-1$ back to 0, we multiply by $\alpha$. Thus each completed cycle of length $m$ contributes one factor $\alpha$.

Set the initial and final vectors as:

$$\lambda = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \quad \text{(start in residue 0)},$$

and

$\rho = (1-\alpha)\, e_c$ where $e_c$ is the standard basis column with a 1 in position $c$.

Equivalently, $\rho$ is the column vector that is $1-\alpha$ at entry $c$ and 0 elsewhere.

## Verification

Let $n \geq 0$. Consider $\mathrm{wt}(a^n) = \lambda\, \mu(a)^n\, \rho$.

**Step 1: If $n \not\equiv c \pmod{m}$, then $\mathrm{wt}(a^n) = 0$.**

Starting from state 0, after reading $n$ symbols the only reachable residue is $n \bmod m$. Hence $\lambda\mu(a)^n$ has support only on the coordinate $n \bmod m$. Since $\rho$ has support only on coordinate $c$, the product is 0 unless $n \bmod m = c$.

**Step 2: If $n \equiv c \pmod{m}$, compute the weight.**

Write $n = mk + c$ with $k \geq 0$ and $0 \leq c < m$. While reading the first $c$ symbols, the automaton moves from residue 0 to residue $c$ without wrapping, so it accumulates no $\alpha$ factor.

After that, the remaining $mk$ symbols consist of exactly $k$ complete wraps from $m-1$ to 0. Each wrap contributes a multiplicative factor $\alpha$, so the total multiplicative factor accumulated from transitions is $\alpha^k$.

Finally, the output vector $\rho$ multiplies by $(1-\alpha)$ exactly when we end in residue $c$. Therefore,

$$\mathrm{wt}(a^{mk+c}) = (1-\alpha)\alpha^k.$$

## Conclusion

The weighted automaton constructed above assigns weight $(1-\alpha)\alpha^k$ to exactly the strings $a^{mk+c}$ and assigns weight 0 to all other lengths. Hence it represents $D(\alpha, m, c)$.